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Next item →

1. Which of the following statements accurately describes the application of De Morgan's theorems in Boolean expressions?

1 / 1 point

- ☐ De Morgan's theorems are used to convert Boolean expressions into exclusive OR operations.
- ☐ De Morgan's theorems can only be applied to expressions involving three or more variables.
- ☒ De Morgan's theorems allow the transformation of conjunctions into disjunctions and vice versa by negating each term and switching the operators.
- ☐ De Morgan's theorems state that the negation of a Boolean expression results in the expression itself.

👏 Nice work

Correct! This is a fundamental aspect of De Morgan's theorems, which simplifies complex Boolean expressions.

2. Which of the following Boolean expressions is equivalent to

1 / 1 point

$$A + \overline{A}B$$

when simplified using Boolean laws?

- ☒ $A + B$
- ☐ AB
- ☐ $\overline{A} + B$
- ☐ $AB + \overline{B}$

👏 Nice work

Correct! The expression

$$A + \overline{A}B$$

simplifies to

$$A + B$$

using the Absorption Law.

3. Select all statements that correctly define or describe a Boolean function and its mapping.

1 / 1 point

- ☒ Boolean functions must always yield the same output for the same set of inputs regardless of how they are implemented.

👏 Nice work

Well done! Consistency is key in Boolean functions, irrespective of their form.

- ☐ Boolean functions can only take a maximum of two variables as input.
- ☒ A Boolean function can be expressed using truth tables, algebraic expressions, or logic gates.

👏 Nice work

Correct! Boolean functions are versatile and can be represented in multiple forms.

- ☐ A Boolean function's mapping is defined only when all possible input combinations have been evaluated.
- ☒ A Boolean function maps a set of binary variables to a binary output.

👏 Nice work

Correct! This is a fundamental property of Boolean functions.

4. Which of the following are valid applications of Karnaugh maps in simplifying Boolean functions?

1 / 1 point

- ☐ Converting Boolean functions to binary numbers
- ☐ Designing sequential circuits
- ☒ Minimizing the number of logic gates in a circuit

👏 Nice work

Correct! Karnaugh maps are used to simplify expressions, which can reduce the number of gates needed.

- ☒ Identifying common sub-expressions in complex Boolean functions

👏 Nice work

Right! Karnaugh maps help in spotting common sub-expressions, aiding simplification.

5. Which of the following logic gates can be used to represent the Boolean expression

1 / 1 point

$$\overline{A \cdot B} + \overline{C}$$

?

- ☒ NOR and NOT gates
- ☐ NAND and XOR gates
- ☐ OR and NOT gates
- ☐ AND and OR gates

👏 Nice work

Correct! The Boolean expression can be represented using NOR and NOT gates. Keep exploring how different gates can implement Boolean expressions.